Probability

Principles of Information Quality

INFQ 70303

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Counting

Probability often involved counting things

Some important formulas that every IQ analyst should know

- Sum of the integers from 1 to N
- Combinations of N things taken M at a time
- Total number of pairs of M objects

Sum of the integers from 1 to N

- Formula is

$$1 + 2 + 3 + \cdots + N = \frac{N \cdot (N + 1)}{2}$$

- Example

$$1 + 2 + 3 + \cdots + 10 = \frac{10 \cdot (10 + 1)}{2} = 55$$

Combinations

- The total number of (unordered) combinations of N things taken M at a time is given by the formula

$$C(N,M) = \frac{N!}{M! \cdot (N-M)!}$$

- When dealing with factorials, keep in mind by definition

$$\mathbf{0! = 1}$$

Combination Example

- How many ways can a committee of 3 people be selected from a group of 10 people.

$$C(10,3) = \frac{10!}{3! \cdot (10-3)!} = \frac{10!}{3! \cdot 7!} = \frac{10 \cdot 9 \cdot 8}{3 \cdot 2 \cdot 1} = 5 \cdot 3 \cdot 8 = 120$$

- Different if Order is Important! For Example:
  How many ways to assign 3 drivers to 10 race cars?
  First driver, 10 choices of car
  Second driver, 9 choices of car
  Third driver, 8 choices of car
  Total = 10x9x8 = 720

Pairs of Objects

- Special Case of Combinations where M = 2
- $Pairs = C(N,2) = \frac{N!}{2!\cdot(N-2)!} = \frac{N\cdot(N-1)}{2}$

- Example: How many ways to compare 10 records?
- Answer: Since order of comparison is not important
-

$$Pairs = C(10,2) = \frac{10 \cdot 9}{2} = 5 \cdot 9 = 45$$

Probability

- **Outcome space** - The set of all possible results of an event, represented as **Ω**.
  - Event: Rolling a pair dice
  - Outcomes: 36 (1-6 first die, 1-6 second die)

- The notion of **probability** is associated with the relative frequency of a certain set of events in the outcome space.
  - Probability dice total 2 = 1/36 = 0.0278 (only 1 event)
  - Probability dice total 4 = 3/36 = 0.0833 (3 events)

Logical Propositions

- **Logical proposition** - a statement describing a condition or event that is true or false.

- **Probability** - is a number between 0 and 1 attached to a logical proposition that represents the likelihood that a proposition will be true.

- We will use the notation P(A) to express the probability of a certain logical expression A.

Basic Axioms of Probability

- The notation P(A) to express the probability of a certain logical expression A.
  - $0 \leq P(A) \leq 1$
  - $P(A) = 1$ if and only if A is certain
  - $P(\sim A) = 1 - P(A)$

- Sum rule: given two logical statements A and B.
  - P(A or B) = P(A) + P(B) - P(A and B)
  - P(A or B) = P(A) + P(B)
    if and only if A and B are mutually exclusive (disjoint)
    i.e. Event A does not influence Event B and vice versa

Conditional Probability

- The probability of occurrence of event A may be influenced by another event B, and this is expressed in terms of a conditional probability.
  - **P(A|B)**
    "Probability of Event A occurring given that Event B occurred"

Product Rule:

- Given two logical statements A and B,
  - **$P(A \text{ and } B) = P(A|B)*P(B) = P(B|A)*P(A)$**
- For "n" logical statements $A_1, A_2, \ldots, A_n$
  - **$P(A_1,A_2,\ldots A_n)=P(A_1)*P(A_2|A_1)*\ldots P(A_n|A_{n-1},\ldots, A_1)$**

Example

Probability of drawing two aces from a deck of 52 cards

- A = Probability of drawing an ace (4/52)
- P(B|A) = Probability of drawing second ace given that the first card was an ace (3/51)
- P(A and B) = Probability of drawing two aces from the same deck
- P(A and B) = P(A) * P(B|A) = (4/52)*(3/51) =.0045 or 0.45% chance

Independent Events

- From the Product Rule
  - **P(A|B) = P(A and B)/P(B)**

- If event A and B are independent
  - **P(A and B) = P(A) * P(B)**
  - **P(A|B) = P(A)**

Question

- What is the probability of drawing a pair of Jacks or better (J, Q, K, A) from a deck of 52 cards?

Answer

- P(A), Probability first card Jack or better = 16/52
- P(B | A), Probability second card matches = 3/51
- P(A and B) = (16/52) * (3/51) = 0.018 or 1.8%

Bayes Theorem

- **Bayes'Theorem** is a way to update probabilities given new evidence.

  If P(H) is the old (prior) hypothesis,

  and D is new data or evidence,

  Then the new probability (posterior hypothesis) is calculated by multiplying the prior hypothesis by a factor, i.e.

  $$\mathbf{P(H|D) = Factor * P(H)}$$

The Factor

$$\text{Factor} = \frac{P(D \mid H)}{P(D \mid H) \cdot P(H) + P(D \mid \sim H) \cdot P(\sim H)}$$

Bayes Theorem

$$P(H \mid D) = \text{Factor} \cdot P(H) = \frac{P(D \mid H) \cdot P(H)}{P(D \mid H) \cdot P(H) + P(D \mid \sim H) \cdot P(\sim H)}$$

Bayes Formula Terms

- **P(H | D)** – new (posterior) probability of a certain hypothesis based on a set of observational data given a certain context. The thing we are looking for
- **P(H)** – the probability the hypothesis is present
- **P(D | H)** – the probability of the data given the hypothesis is present
- **P(D | ~H)** – the probability of the data given the hypothesis is not present
- **P(~H)** – the probability the hypothesis is not present

Example

- The probability that someone the population has an abnormal gene G is 0.1%
- A new test for gene G has been developed that has 3% false positive, and a 2% false negative rate.
- If someone take the test that shows positive, what is the probability that he/she has gene G?

Solution

- P(H) probability of having gene G is 0.001
- P(~H) probability of not having gene G is .999
- D is testing positive for gene G
- P(D | H) probability of positive if you have gene G is .98
- P(D | ~H) probability positive if you don't have gene G is .03
- $P(H|D) = \frac{P(D|H)\cdot P(H)}{P(D|H)\cdot P(H)+P(D|\sim H)\cdot P(\sim H)}$
- P(H | D) = (.98*.001)/((.98)(.001)+(.03)(.999)) = .031
  Only 3% probability you have gene G even if you test positive for it!

Another View of Bayes

- Fallacy that P(H | D) = P(D | H)

- This is NOT ALWAYS TRUE
  – it depends upon the ratio of P(H) and P(D).

- The correct relationship is

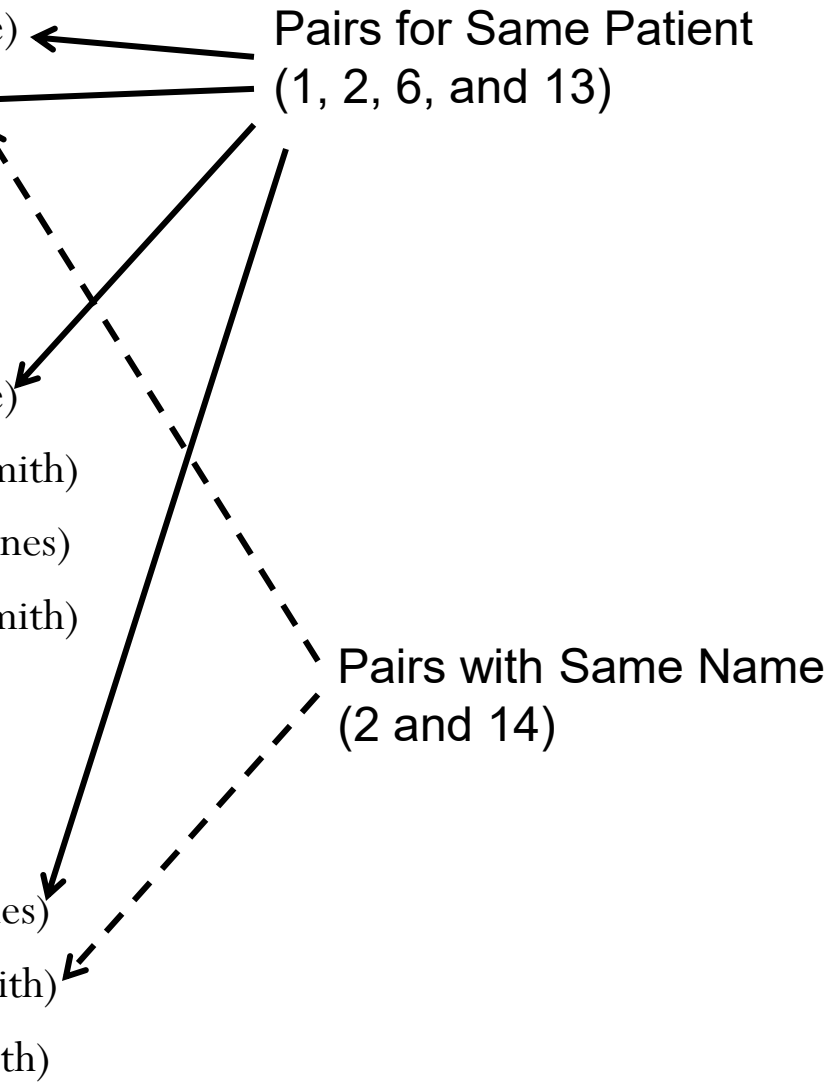
$$P(H|D) = \frac{P(H)}{P(D)} \cdot P(D|H)$$

Example: Six Patient Records

- Patient 1
  - **r1** =Bill Doe, **r2**=William Doe, **r3**=Bill Doe
- Patient 2
  - **r4**=Mary Smith, **r5**=Mary Jones
- Patient 3
  - **r6**=Mary Smith

There are $\frac{6 \cdot (6-1)}{2} = 15$ ways to compare the 6 records

1. (p1:r1:Bill Doe, p1:r2:William Doe)
2. (p1:r1:Bill Doe, p1:r3:Bill Doe)
3. (p1:r1:Bill Doe, p2:r4:Mary Smith)
4. (p1:r1:Bill Doe, p2:r5:Mary Jones)
5. (p1:r1:Bill Doe, p3:r6:Mary Smith)
6. (p1:r2:William Doe, p1:r3:Bill Doe)
7. (p1:r2:William Doe, p2:r4:Mary Smith)
8. (p1:r2:William Doe, p2:r5:Mary Jones)
9. (p1:r2:William Doe, p3:r6:Mary Smith)
10. (p1:r3:Bill Doe, p2:r4:Mary Smith)
11. (p1:r3:Bill Doe, p2: r5:Mary Jones)
12. (p1:r3:Bill Doe, p3:r6:Mary Smith)
13. (p2:r4:Mary Smith, p2:r5:Mary Jones)
14. (p2:r4:Mary Smith, p3:r6:Mary Smith)
15. (p2:r5:Mary Jones, p3:r6:Mary Smith)



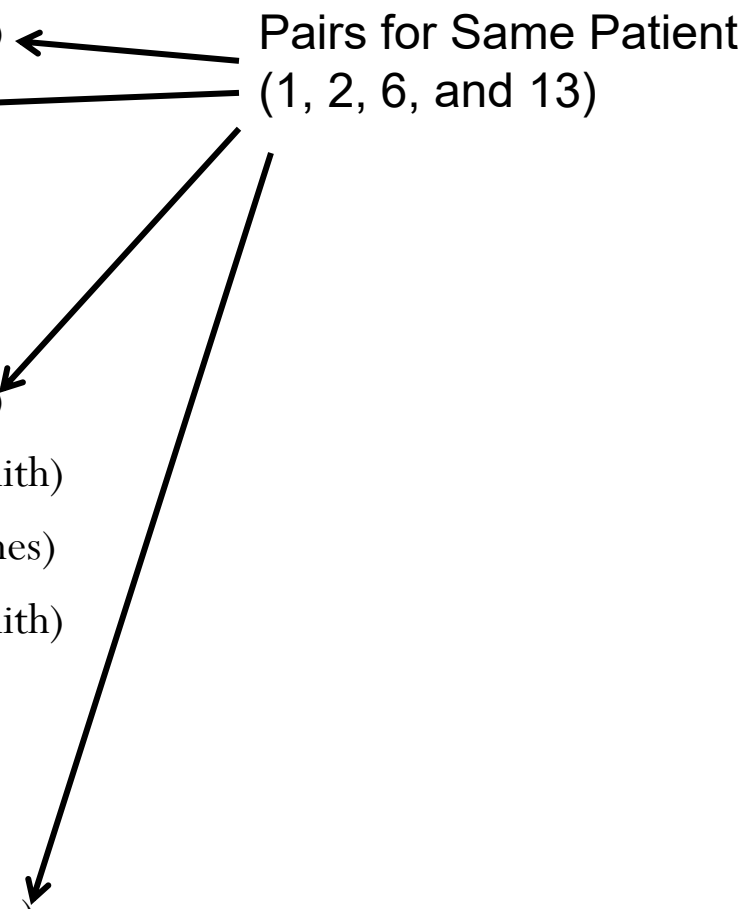
Questions

If a pair of records is selected at random,

- What is P(sp)? {will be for same patient}
- What is P(sn)? {will have the same name|
- What is P(sp|sn)? {for same patient if have same name}
- What is P(sn|sp)? {have same name if for same patient}

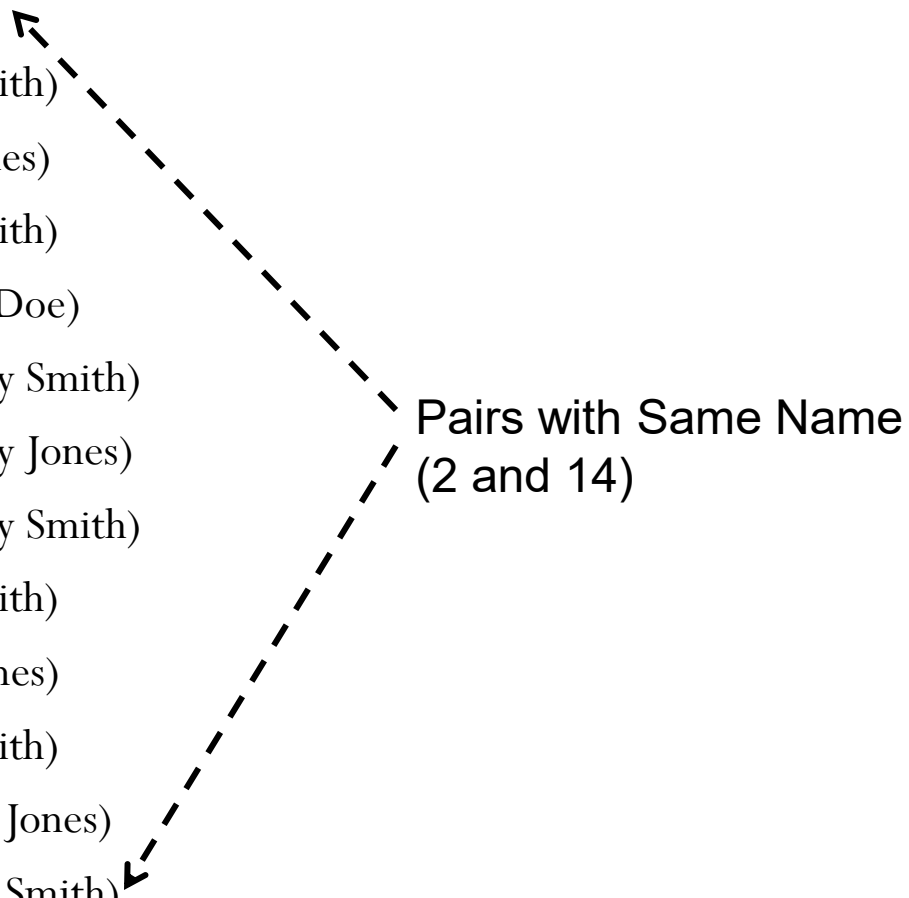
P(sp) = 4/15

1. (p1:r1:Bill Doe, p1:r2:William Doe)
2. (p1:r1:Bill Doe, p1:r3:Bill Doe)
3. (p1:r1:Bill Doe, p2:r4:Mary Smith)
4. (p1:r1:Bill Doe, p2:r5:Mary Jones)
5. (p1:r1:Bill Doe, p3:r6:Mary Smith)
6. (p1:r2:William Doe, p1:r3:Bill Doe)
7. (p1:r2:William Doe, p2:r4:Mary Smith)
8. (p1:r2:William Doe, p2:r5:Mary Jones)
9. (p1:r2:William Doe, p3:r6:Mary Smith)
10. (p1:r3:Bill Doe, p2:r4:Mary Smith)
11. (p1:r3:Bill Doe, p2: r5:Mary Jones)
12. (p1:r3:Bill Doe, p3:r6:Mary Smith)
13. (p2:r4:Mary Smith, p2:r5:Mary Jones)
14. (p2:r4:Mary Smith, p3:r6:Mary Smith)
15. (p2:r5:Mary Jones, p3:r6:Mary Smith)



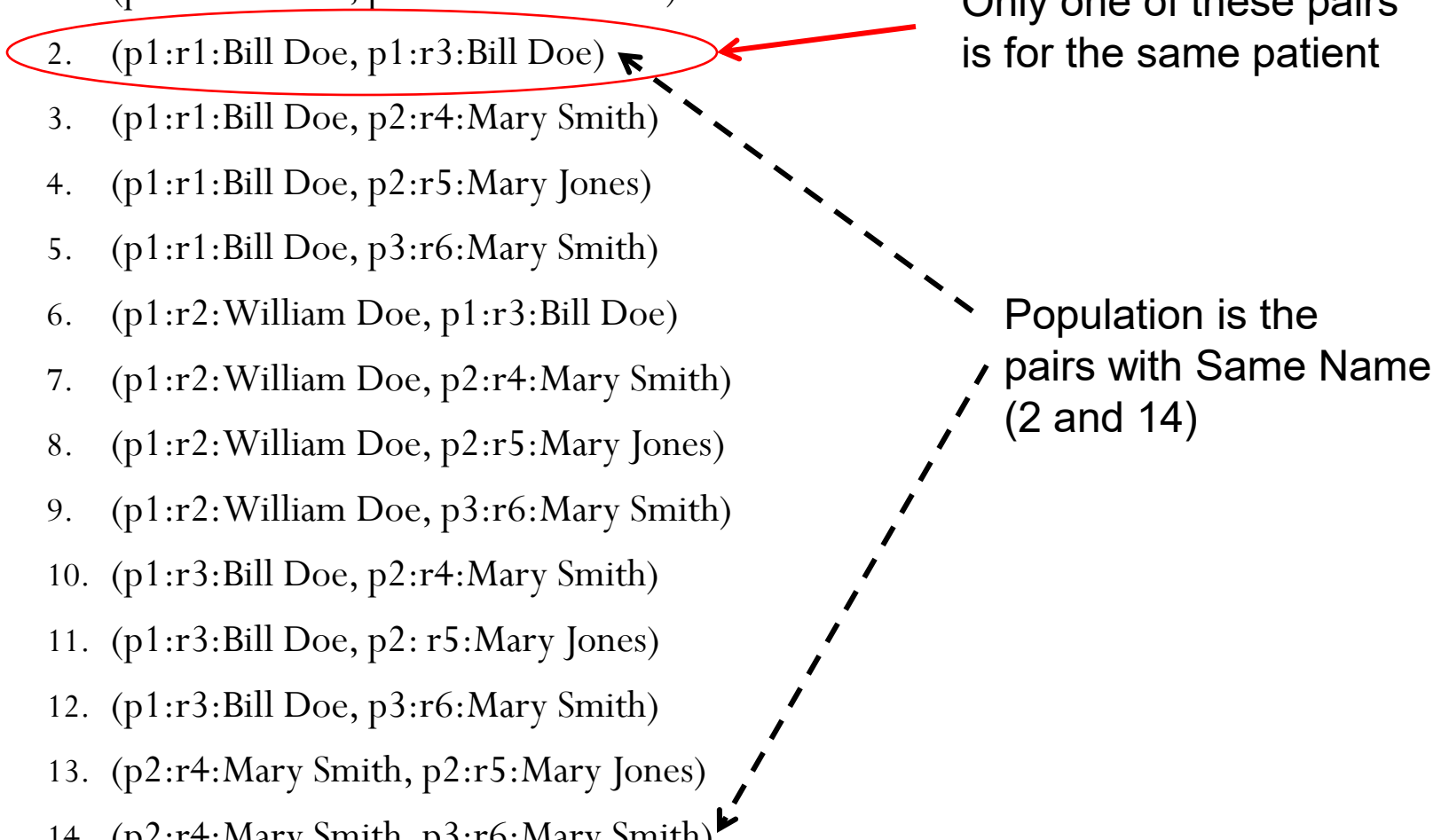
P(sn) = 2/15

1. (p1:r1:Bill Doe, p1:r2:William Doe)
2. (p1:r1:Bill Doe, p1:r3:Bill Doe)
3. (p1:r1:Bill Doe, p2:r4:Mary Smith)
4. (p1:r1:Bill Doe, p2:r5:Mary Jones)
5. (p1:r1:Bill Doe, p3:r6:Mary Smith)
6. (p1:r2:William Doe, p1:r3:Bill Doe)
7. (p1:r2:William Doe, p2:r4:Mary Smith)
8. (p1:r2:William Doe, p2:r5:Mary Jones)
9. (p1:r2:William Doe, p3:r6:Mary Smith)
10. (p1:r3:Bill Doe, p2:r4:Mary Smith)
11. (p1:r3:Bill Doe, p2: r5:Mary Jones)
12. (p1:r3:Bill Doe, p3:r6:Mary Smith)
13. (p2:r4:Mary Smith, p2:r5:Mary Jones)
14. (p2:r4:Mary Smith, p3:r6:Mary Smith)
15. (p2:r5:Mary Jones, p3:r6:Mary Smith)



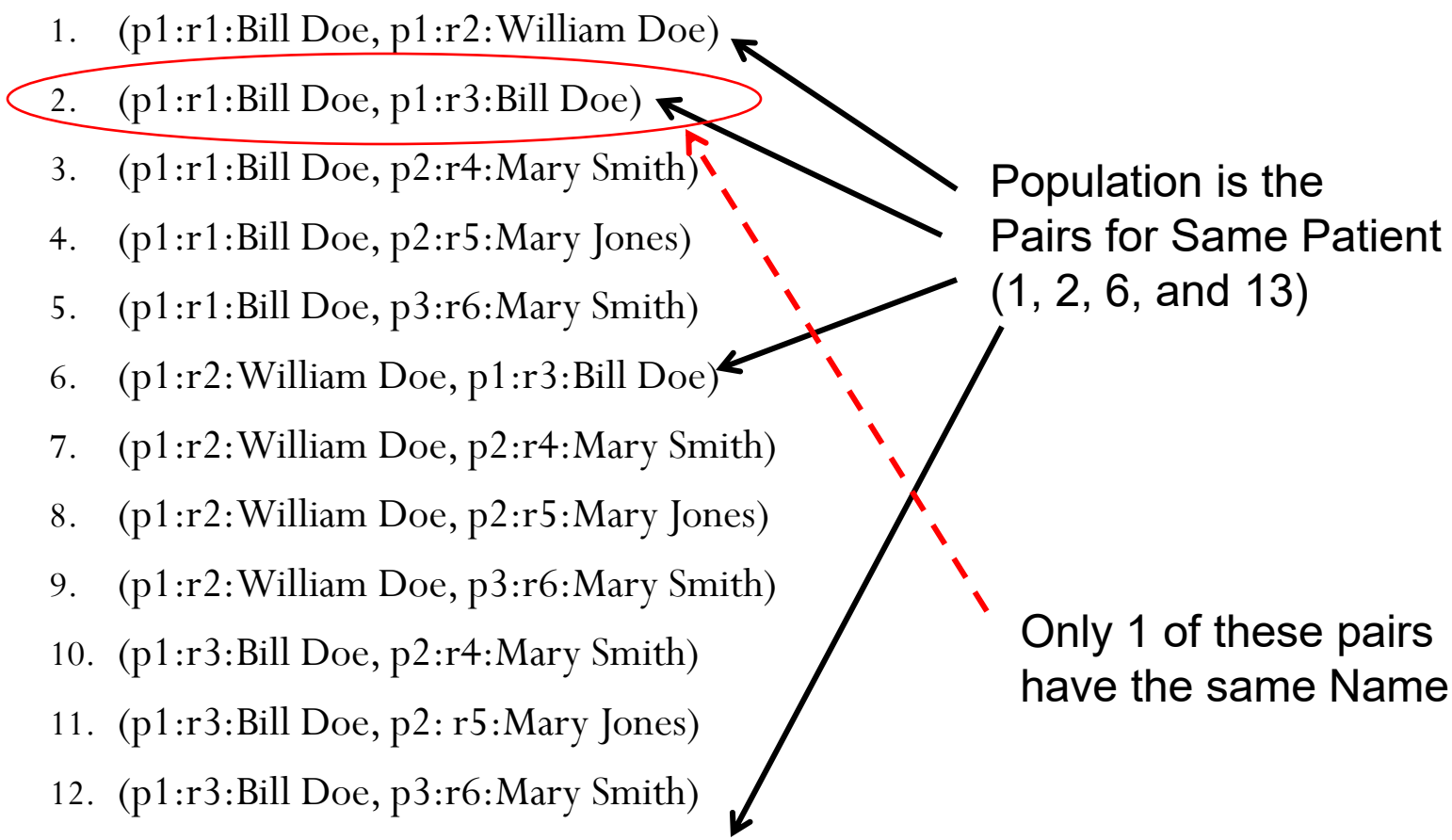
P(sp|sn) = 1/2

1. (p1:r1:Bill Doe, p1:r2:William Doe)
2. (p1:r1:Bill Doe, p1:r3:Bill Doe)
3. (p1:r1:Bill Doe, p2:r4:Mary Smith)
4. (p1:r1:Bill Doe, p2:r5:Mary Jones)
5. (p1:r1:Bill Doe, p3:r6:Mary Smith)
6. (p1:r2:William Doe, p1:r3:Bill Doe)
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13. (p2:r4:Mary Smith, p2:r5:Mary Jones)
14. (p2:r4:Mary Smith, p3:r6:Mary Smith)
15. (p2:r5:Mary Jones, p3:r6:Mary Smith)



P(sn|sp) = 1/4

1. (p1:r1:Bill Doe, p1:r2:William Doe)
2. (p1:r1:Bill Doe, p1:r3:Bill Doe)
3. (p1:r1:Bill Doe, p2:r4:Mary Smith)
4. (p1:r1:Bill Doe, p2:r5:Mary Jones)
5. (p1:r1:Bill Doe, p3:r6:Mary Smith)
6. (p1:r2:William Doe, p1:r3:Bill Doe)
7. (p1:r2:William Doe, p2:r4:Mary Smith)
8. (p1:r2:William Doe, p2:r5:Mary Jones)
9. (p1:r2:William Doe, p3:r6:Mary Smith)
10. (p1:r3:Bill Doe, p2:r4:Mary Smith)
11. (p1:r3:Bill Doe, p2: r5:Mary Jones)
12. (p1:r3:Bill Doe, p3:r6:Mary Smith)
13. (p2:r4:Mary Smith, p2:r5:Mary Jones)
14. (p2:r4:Mary Smith, p3:r6:Mary Smith)
15. (p2:r5:Mary Jones, p3:r6:Mary Smith)



Probabilities

- P(sp) = 4/15
- P(sn) = 2/15
- P(sp | sn) = ½
- P(sn | sp) = ¼
- Bayes

  P(sp | sn) = (P(sp)/P(sn))x P(sn | sp)

  = (4/15) x (15/2) x ¼ = (4/2) x ¼ = 2 x ¼ = ½

Random Variables and Probability Distributions

Random Variables

- A random variable X is a function that maps the outcome space Ω to the set of the real numbers.

- It can be viewed as the name of an event with a probabilistic outcome.

Discrete Distributions

- A probability distribution function ***P*** for a random variable X determines the probability that the random variable X will take the value v.
  It is denoted as

$$\mathbf{p(v) = P(X=v).}$$

- If the random variable is **discrete**, from the axioms of probability it follows that

$$1 = \sum_{x} P(X = v)$$

Example

- X is probability of the total of a two die being thrown. The possible totals are 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12
- $p(2) = P(X=2) = 1/36 = p(12) = P(X=12)$
- $p(3) = P(X=3) = 2/36 = p(11) = P(X=11)$
- $p(4) = P(X=4) = 3/36 = p(10) = P(X=10)$
- $p(5) = P(X=5) = 4/36 = p(9) = P(X=9)$
- $p(6) = P(X=6) = 5/36 = p(8) = P(X=8)$
- $p(7) = P(X=7) = 6/36$
- $1/36+2/36+3/36+4/36+5/36+6/36+5/36+4/36+3/36+2/36+1/36 = 36/36=1$

Binomial Distribution

- p = probability of success on a trial
- n = number of trials
- s = number of successes

$$P(X = s|p, n) = \frac{n!}{s! \cdot (n-s)!} \cdot p^s \cdot (1-p)^{n-s}$$

Coin Flip

- **p**(heads)=0.5, **n** = 5 flips, **s** = 4 heads
- Probability of exactly 4 heads in 5 flips
- $P(X = s|p, n) = \frac{n!}{s!\cdot(n-s)!} \cdot p^s \cdot (1-p)^{n-s}$
- $P(X = 4|0.5,5) = \frac{5!}{4!\cdot(5-4)!} \cdot 0.5^4 \cdot (1-0.5)^{5-4}$
- $= 5 \cdot 0.0625 \cdot 0.5 = 0.15625$
- BINOM.DIST(4, 5, 0.5, false) = 0.15625
- Probability of 3 heads or less in 5 flips
- BINOM.DIST(3, 5, 0.5, true) = 0.81250
- Probability of 4 heads or more in 5 flips
- 1.0 - BINOM.DIST(3, 5, 0.5, true) = 1.0 – 0.81250 = 0.1875

Density Function

- When random variables are **continuous**, then $P(X = x) = 0$ for every real value.
- We must therefore calculate probabilities within a certain interval of values of X.
- Probabilities are expressed in terms of probability density functions where the density function $f(x)$ is such that :

$$P(a < x < b) = \int_a^b f(x)\,\mathrm{dx}$$

$$f(x) \geq 0, \forall x \in \mathcal{R}$$

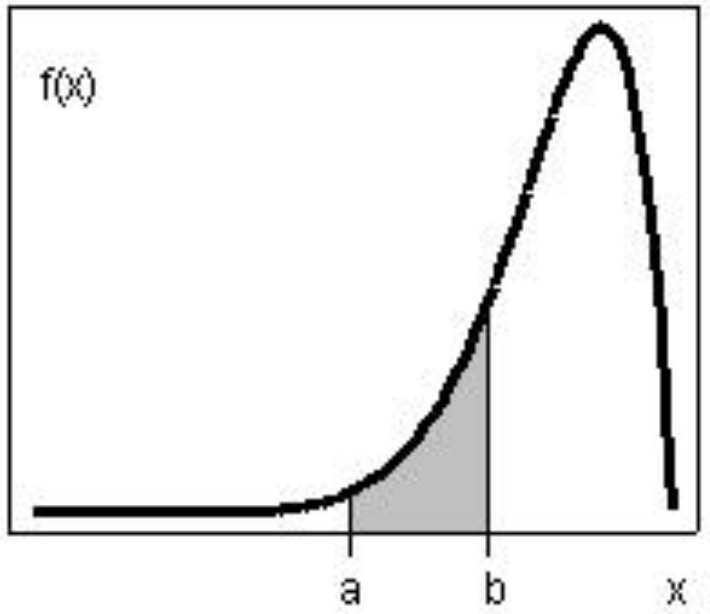
Distribution Function

- The total area under the density function curve is equal to 1, and the area between a and b is equal to $P(a<x<b)$.

- For both continuous and discrete values, the expression $F(x) = P(X \leq x)$ is known as the **distribution function**.

- Its value for a certain value x is the probability that the random variable X will be less than or equal to x.

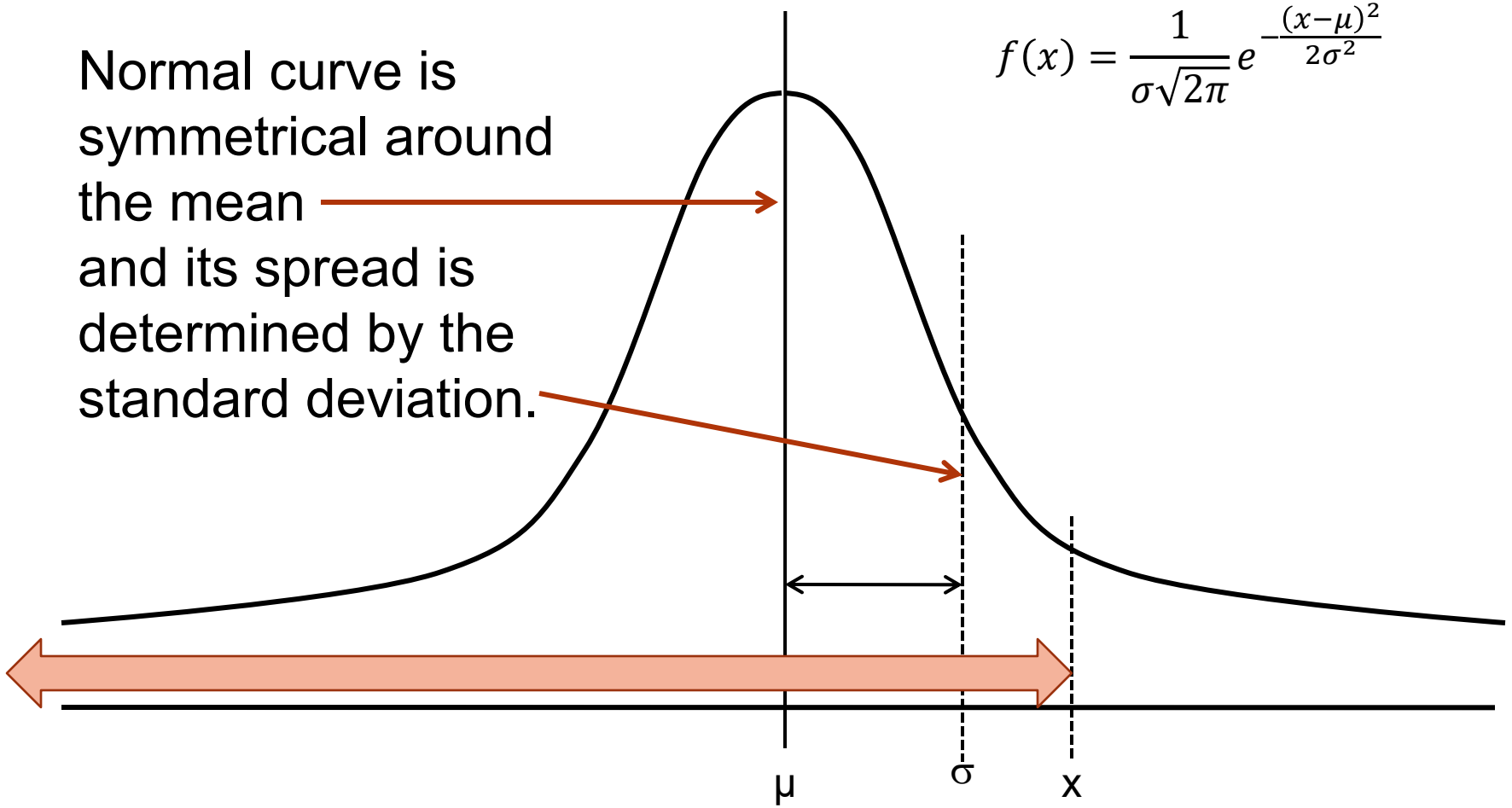
Figure 5.5 Continuous Distribution



Normal (Gaussian) Distribution

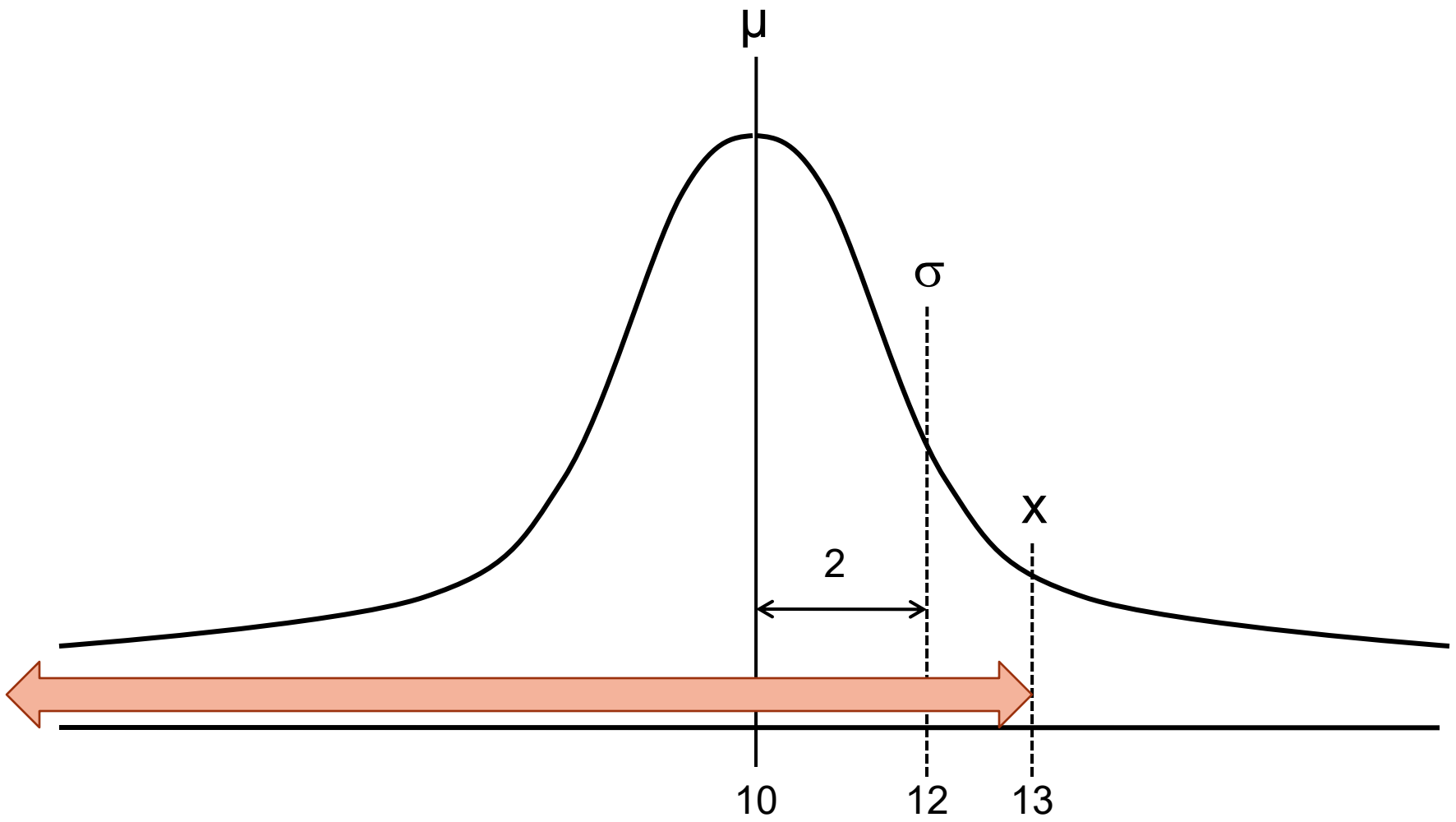
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Normal curve is symmetrical around the mean and its spread is determined by the standard deviation.



NORM.DIST(x, μ, σ,TRUE) give the Probability of values ≤ x

Example Normal Distribution



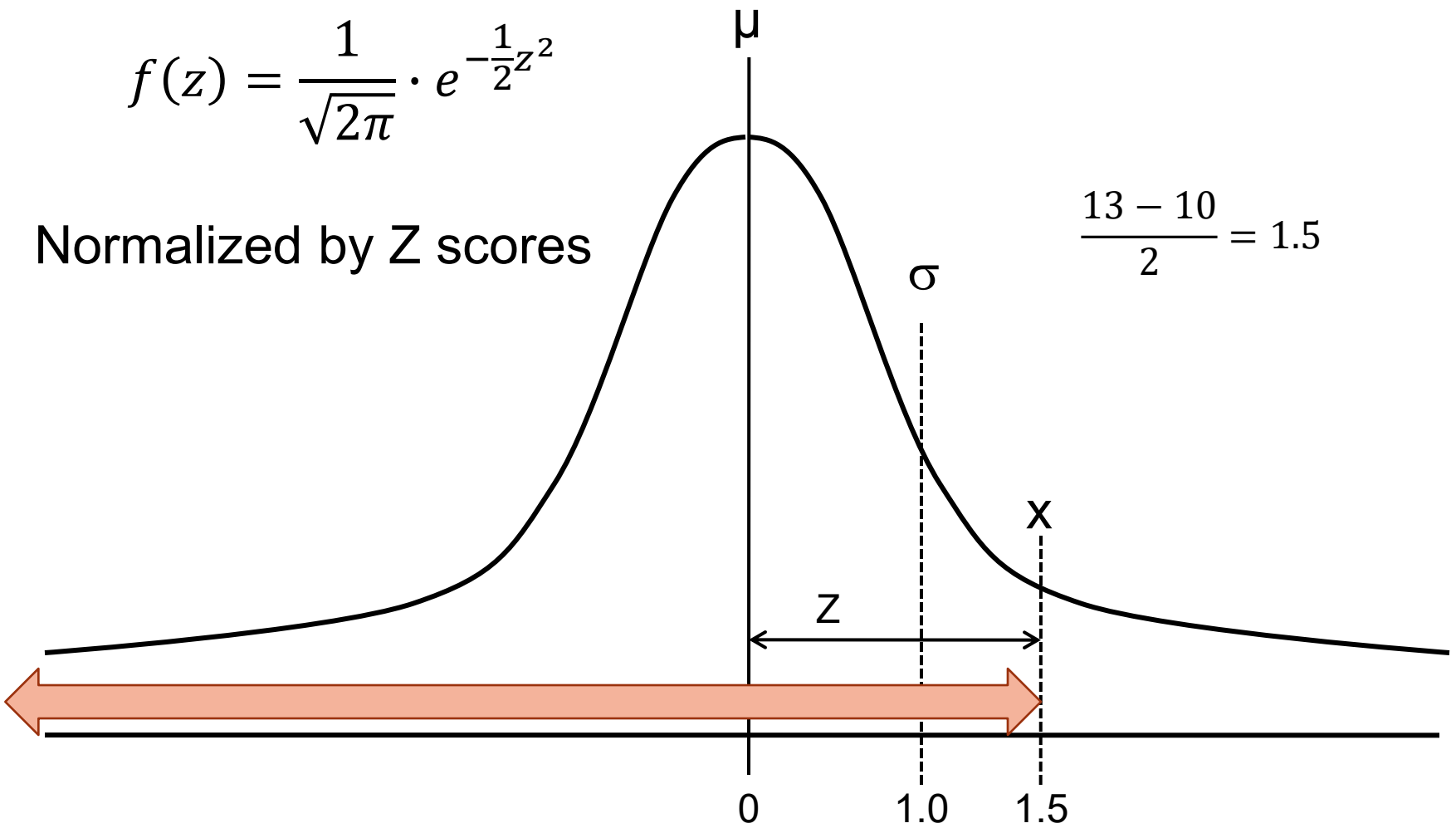
NORM.DIST(13,10, 2,TRUE) = 0.933192 (93.3% probability of values ≤ 13)

Normalized Normal Distribution

$$f(z) = \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{1}{2}z^2}$$

Normalized by Z scores

$$\frac{13 - 10}{2} = 1.5$$

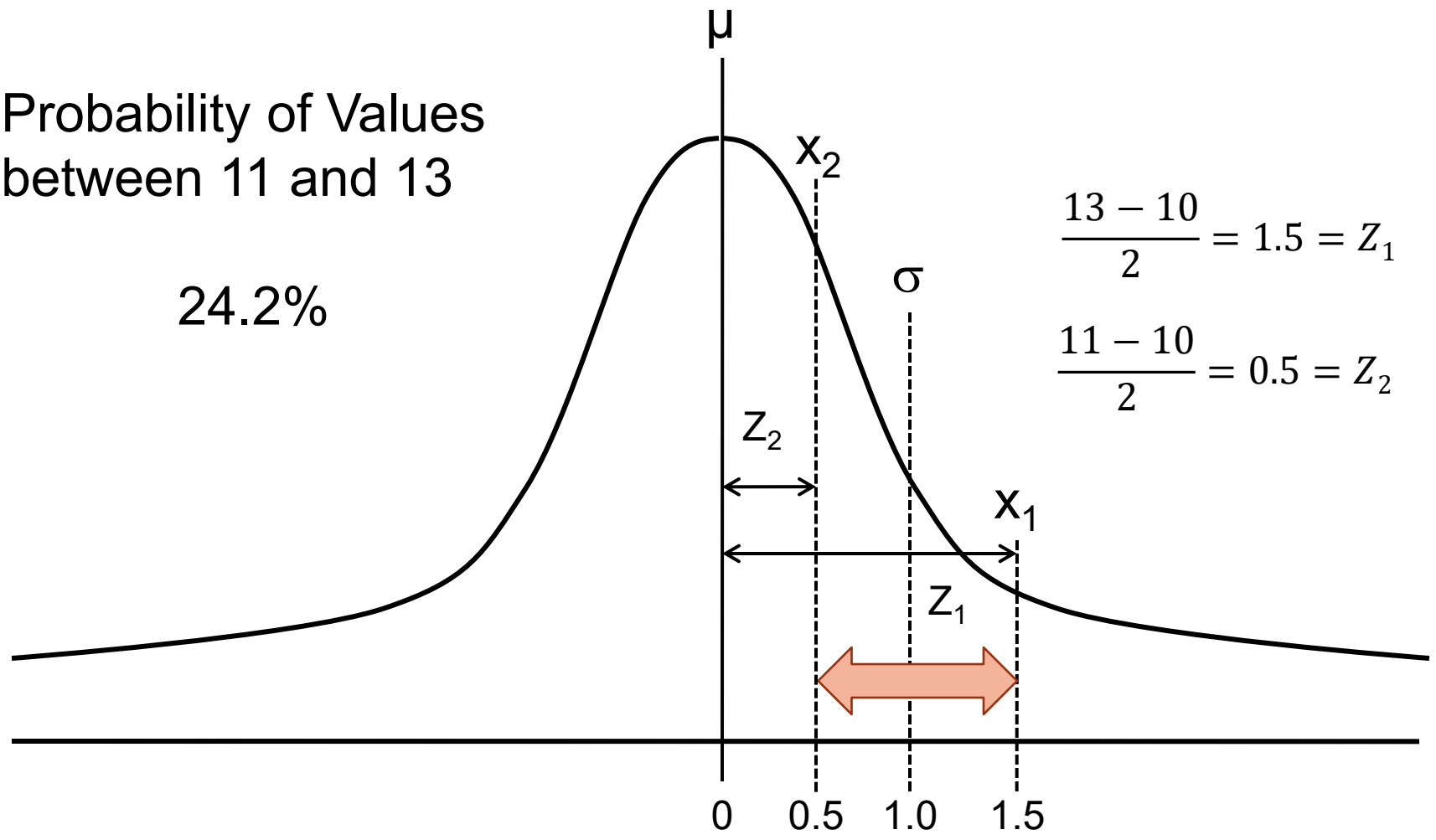


NORM.S.DIST(1.5) = 0.933192 (93.3% probability of Z-scores ≤ 1.5)

Probability of Range of Values

Probability of Values between 11 and 13

24.2%

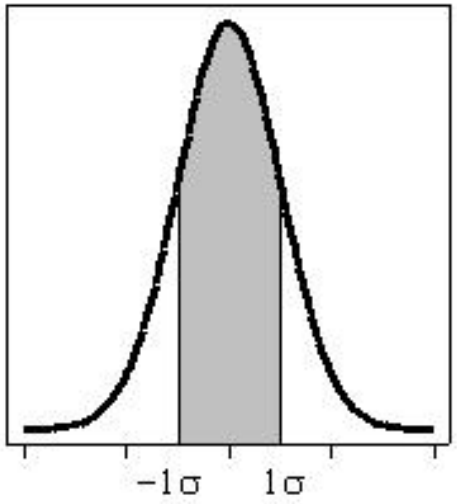


$$\frac{13-10}{2} = 1.5 = Z_1$$

$$\frac{11-10}{2} = 0.5 = Z_2$$

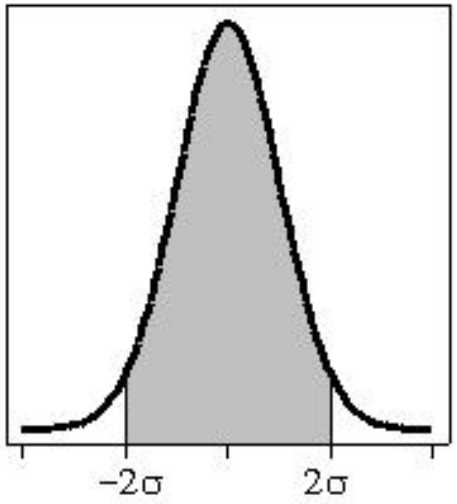
NORM.S.DIST(1.5) – NORM.S.DIST(0.5) = 0.933192 – 0.691462 = 0.241730

Z-Scores in Normal Distributions



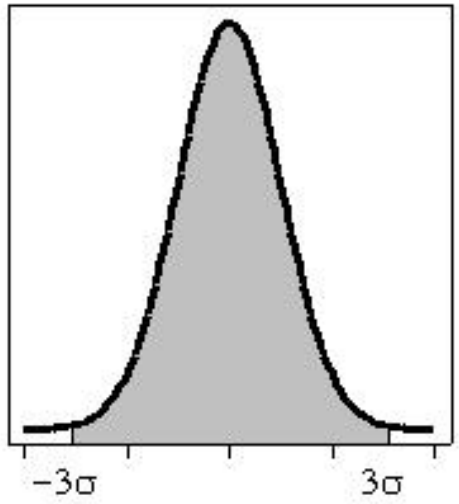
Z (in std. deviation units)

(a) shaded area= 68.27%



Z (in std. deviation units)

(a) shaded area= 95.45%



Z (in std. deviation units)

(a) shaded area= 99.73%

Questions and Discussion